# CH PRASHANTH BABU

HYDERABAD, TELANGANA

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## Education

**Indian Institute of Information Technology Bhagalpur** **Aug 2023 – May 2027**
*B.Tech, Electronics and Communication Engineering* *Bhagalpur, Bihar*

## Experience

**MercuryAI Innovations India Pvt. Ltd.** **Sep 2025 – Oct 2025**
*Frontend Developer Intern* *Remote*

- Developed **12+ reusable React.js and Next.js UI components**, improving page performance by **20%** through component-level optimization.
- Delivered frontend features across **2 Agile sprint cycles**, implementing responsive layouts and reusable components to meet assigned development requirements.
- Refactored frontend modules to reduce code duplication, improve component readability, and maintain consistent UI behavior across application pages.

## Projects

**AI Tour Planner** | Python, LangGraph, Claude, Gemini, Pydantic, Streamlit **May 2026 – Jun 2026**

- Architected a **5-agent AI travel planner** using LangGraph StateGraph and Claude for intent extraction, destination research, itinerary generation, budget estimation, and image generation.
- Integrated **2 REST APIs** (API Ninjas and Wikipedia) for weather, attractions, currency, and destination data, enabling personalized recommendations, automated currency detection, and budget validation.
- Implemented **6+ features**, including Pydantic v2 structured outputs, Gemini image generation, live agent-progress streaming, session history, JSON/PDF exports, and Docker deployment with Streamlit.

**Productionized Multimodal RAG System** | Python, FastAPI, LangChain, Qdrant, PostgreSQL, Redis, Celery

- Engineered a **Multimodal RAG pipeline** supporting **100+ documents** across PDFs, images, and tables, integrating document ingestion, semantic search, hybrid retrieval, reranking, and citation-grounded question answering.
- Built an asynchronous ingestion architecture using FastAPI, Celery, and Redis, integrating Qdrant vector search, PostgreSQL metadata management, document versioning, and multi-tenant access control to process documents at scale.
- Developed an evaluation pipeline with **50+ test questions** targeting **90%+ retrieval accuracy**, alongside OCR fallback, observability, and automated testing; benchmarked retrieval and response latency to target a **20–40% latency reduction**.

## Technical Skills

AI/LLMs: Generative AI, LLMs, Agentic AI, MCP, RAG, Prompt Engineering, LangChain, Embeddings, Vector Databases, Tool Calling, Multi-Agent Systems, OpenAI API, Gemini API, AI-Assisted Development
Languages: C, C++, Java, JavaScript, Python, SQL
Frontend: React.js, Next.js, TypeScript, HTML5, CSS3, Tailwind CSS, Streamlit
Backend: Node.js, Express.js, FastAPI, REST APIs, Authentication (JWT)
Databases and Messaging: MongoDB, PostgreSQL, MySQL, Qdrant, Redis, Celery
Cloud/DevOps: AWS, Docker, Git, GitHub, CI/CD, Vercel, Cloud Deployment
Software Engineering: SDLC, System Design, Distributed Systems, Scalable Applications, Cloud Architecture, Software Testing, Automated Testing, Debugging, Troubleshooting, Continuous Improvement
Tools: Stripe, Mapbox, Postman, VS Code
Coursework: DSA, DBMS, Operating Systems, Computer Networks, OOPS

## Leadership / Achievements

- **Competitive Programming:** Solved **500+ coding problems** across LeetCode (**300+**) and GeeksforGeeks (**200+**), with regular participation in competitive programming contests.
- Completed **Full-Stack Web Development** course at Apna College.
- **Oracle Cloud Infrastructure 2025 Certified Developer Professional** – Certified by Oracle for advanced cloud-native development, OCI service integration, and scalable application deployment.
- **ServiceNow Certifications:** Earned **ServiceNow CSA** and **CAD** certifications with **80%+ scores**, validating expertise in ServiceNow platform administration, application development, and workflow automation.